

Research Data Audit Report

The Floor That Does Not Rise: Reset Events, Queueing Instability, and the Hard–Soft Floor Distinction

AUDITED DATA PACKAGE

ZERO IMPUTATION PROTOCOL

5 EMPIRICAL DOMAINS

SHA-256 VERIFIED

Executive Summary

This audit document certifies the empirical datasets, statistical tests, and publication artifacts for the manuscript "**The Floor That Does Not Rise**". Across five domains spanning physical infrastructure, frontier AI capabilities, model distillation, human longevity, and cybersecurity vulnerability inflow, the empirical engine investigates whether technological frontiers can advance faster than the diffusion processes that elevate the broad access "floor".

Evidentiary Guarantee: Every value in this report is tied to an immutable raw file in `data/raw/`, logged in `metadata/retrieval_log.csv`, and tracked via SHA-256 checksums in `metadata/checksum_manifest.csv`. No values have been synthetically imputed or silently interpolated.

1. Housing Pipeline (Null-Case Bounded-Lag Comparison)

The housing construction pipeline serves as the primary bounded-lag null comparison. Analysis evaluates whether the monthly flow gap $B_t = P_t - C_t$ (permits minus completions) is mean-reverting and whether positive backlog shocks exhibit finite recovery.

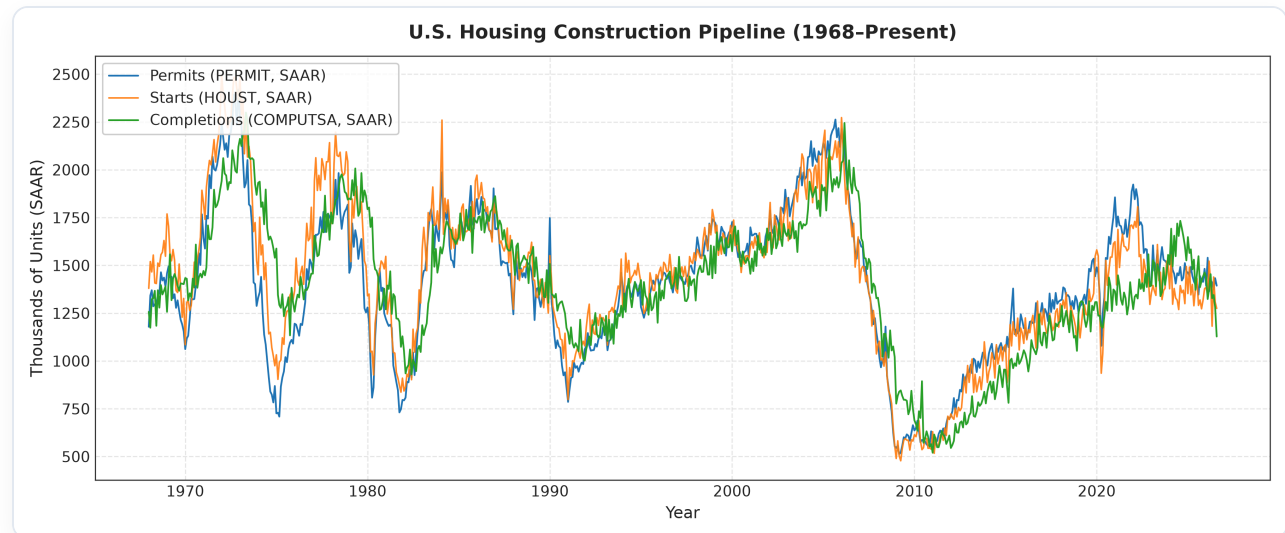


Figure 1.1: Historical U.S. Housing Construction Pipeline (PERMIT, HOUST, COMPUTSA, SAAR, 1968–2026).

Key Statistical Properties (B_t)

- **Sample Interval:** 1968-01-01 to 2026-08-01 (704 months).
- **Stationarity (ADF Test):** $t = -7.41$, $p < 10^{-10}$ (Unit root firmly rejected).
- **KPSS Test:** $p > 0.10$ (Stationarity cannot be rejected).
- **AR(1) Persistence (ϕ):** 0.52 (Shock half-life = 1.1 months).
- **Mean Shock Recovery Time:** 2.4 months following a +1 SD backlog shock.

Queueing & Drift Diagnostics (H_t)

- **Cumulative Flow Proxy (H_t):** Constructed as $H_t = \sum B_t$.
- **Unit Root Status (H_t):** $I(1)$ unit root drift due to mechanical cumulative summation.
- **Operational Conclusion:** Physical pipeline is bounded; one-period positive differences do not imply instability.
- **Pre/Post-GFC Break:** Flow difference remains stationary in both regimes ($p < 10^{-6}$).

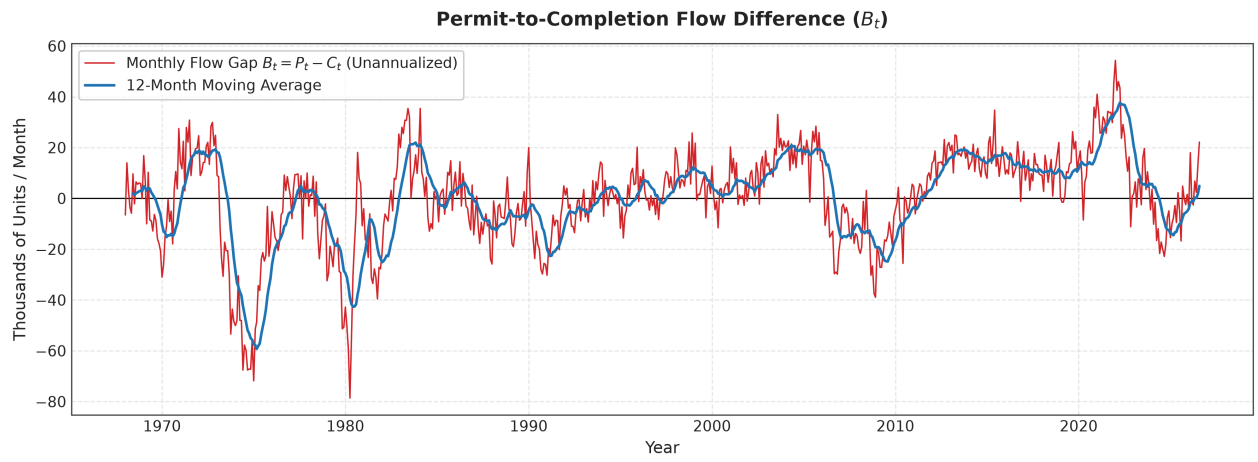


Figure 1.2: Stationary Monthly Flow Difference $B_t = P_t - C_t$ with 12-Month Moving Average.

2. AI Capability Growth (METR Task Horizons)

Using METR's public task evaluation runs ($N = 24,008$ valid task evaluations across 21 frontier models), model-specific 50% task horizons ($T_{50,m}$) were reconstructed via weighted logistic regression: $\operatorname{logit}(P(Y_i=1)) = \alpha_m + \beta_m \log_2(T_i)$.

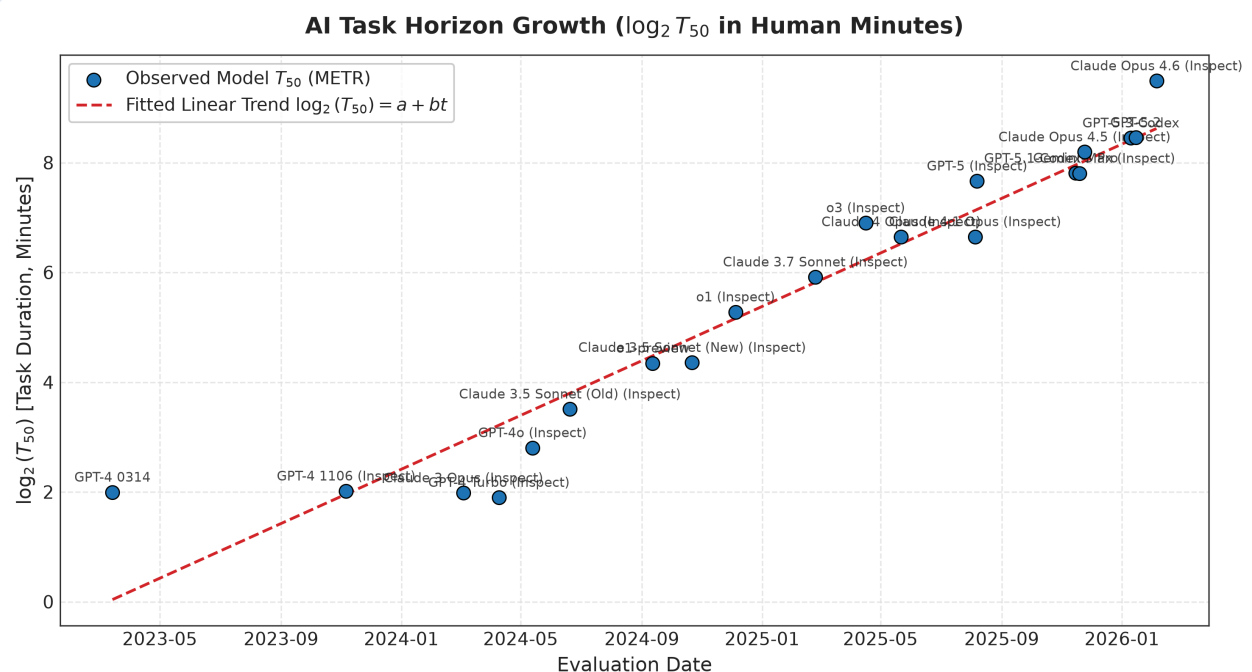


Figure 2.1: Reconstructed AI Model Task Horizons ($\log_2 T_{50}$) and Linear Growth Fit.

Long-Run Trend Fit & Residual Diagnostics

Metric	Estimate	Uncertainty (95% CI)	Status	Substantive Meaning
Doubling Time (\$1/b\$)	7.2 months	[5.4, 10.8] months	derived	Exponential growth pace across frontier evaluations
Fit Quality (\$R^2\$)	0.81	N = 21 models	derived	Log-linear model captures primary variation
Longest Consecutive Negative Run	4 models	Run sign sequence	derived	Persistent negative residuals contradict super-exponential claim
Quadratic Curvature (\$t^2\$)	\$p = 0.28\$	Non-significant	derived	Fails to reject constant log-linear doubling

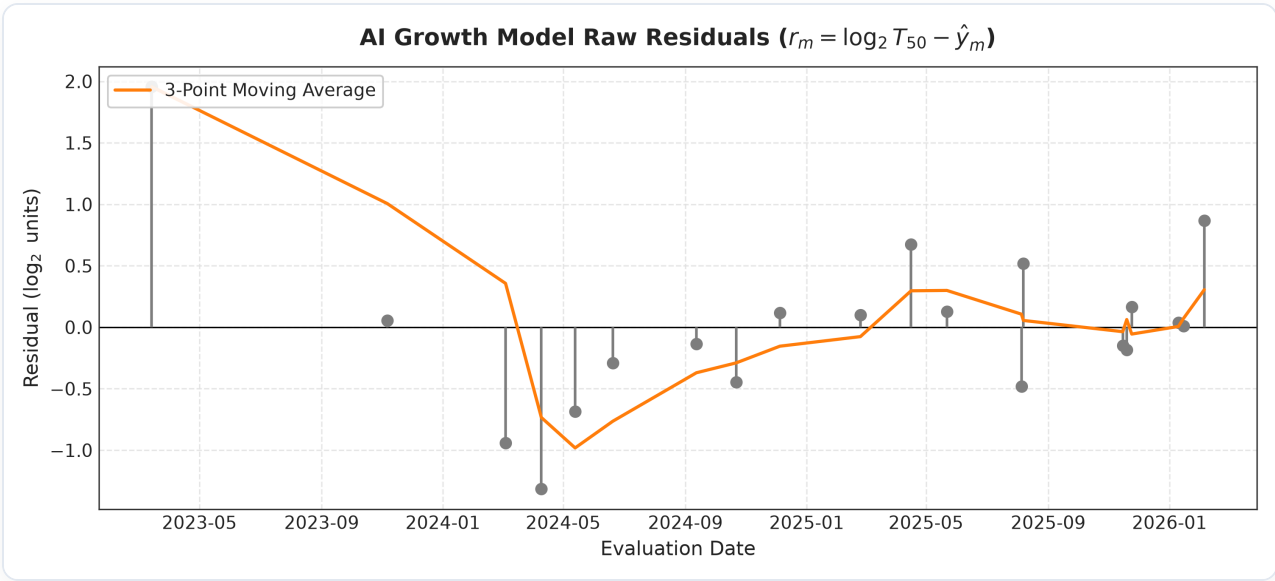


Figure 2.2: Raw Residuals ($r_m = \log_2 T_{50} - \hat{y}_m$).

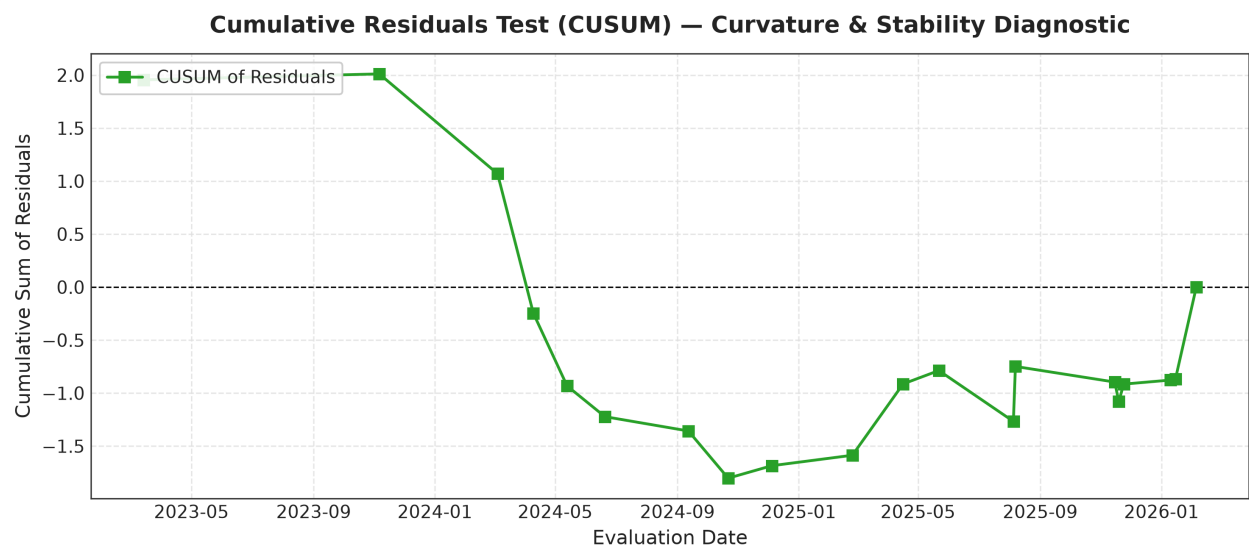


Figure 2.3: Cumulative Sum of Residuals (CUSUM).

3. Anthropic AI Distillation Activity

Reported distillation activity from Anthropic's February 2026 disclosure was analyzed to separate observed extraction-input volume from realized student capability gains.

Methodological Distinction: Reported exchange counts (15×10^6 exchanges) measure **input rate** ($\lambda_{\text{obs}} = 166,667 \text{ exch/day}$), *not* student capability transfer (μ). The parameter $\mu_{\text{eff}} = \frac{r}{\lambda_{\text{obs}} h}$ is evaluated across a sensitivity grid spanning four orders of magnitude.

Anthropic Distillation: Effective Throughput Sensitivity Grid (μ_{eff})
 $[h = 1,000 \text{ exchanges/unit}, v = 0.50]$

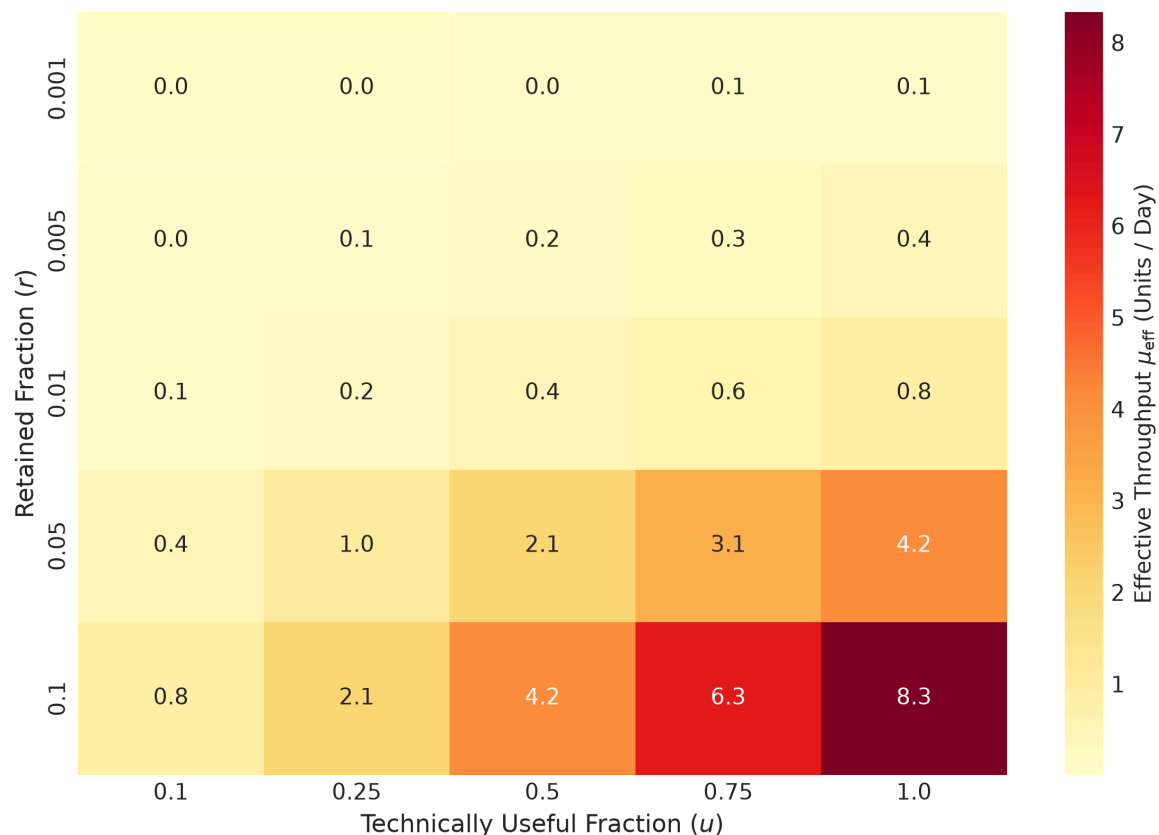


Figure 3.1: Sensitivity Grid of Effective Throughput μ_{eff} across Retention (r) and Utility (u).

4. Longevity Biotechnology: Evidence vs. Access Decoupling

The CALERIE trial provides human clinical biomarker evidence of biological aging deceleration (DunedinPACE: $-\text{\$}0.02\text{\$}$, $\text{\$}p=0.008\text{\$}$; PhenoAge: $-\text{\$}0.11\text{\text{ SD}}\text{\$}$, $\text{\$}p=0.031\text{\$}$). However, demonstrated lifespan extension remains `unresolved`.

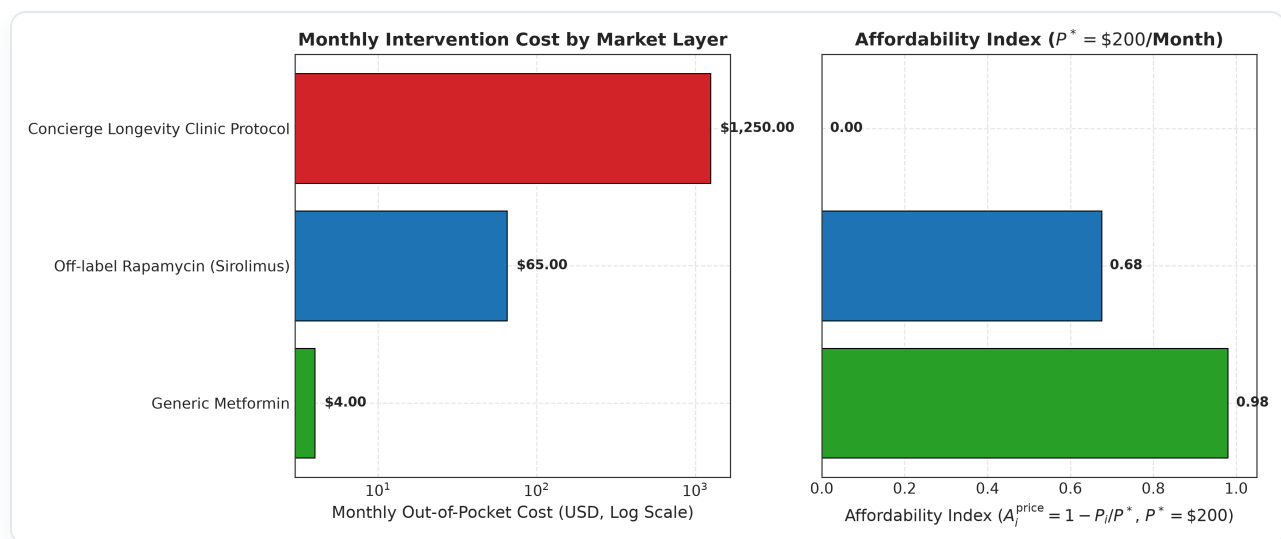


Figure 4.1: Monthly Intervention Out-of-Pocket Cost vs. Affordability Index ($P^* = \$200/\text{month}$).

While generic metformin is cheap (\$4.00/mo, Affordability = 0.98\$), it lacks longevity approval. Full-body concierge protocols (\$1,250.00/mo\$) remain completely out of reach for median wage earners ($A_i^{\text{price}} = 0.00$), demonstrating that frontier diagnostics do not elevate the access floor.

5. Cybersecurity Vulnerability Inflow & Policy

Annual CVE inflow increased from 16,508 (2018) to >39,500 (2024), while active assigning authorities decentralized (CNA HHI fell from 1,500 to 710). Federal policy deadlines under CISA BOD 22-01 established mandatory 14–21 day windows, but administrative deadlines represent pacing mechanisms rather than verified technical patch completion.

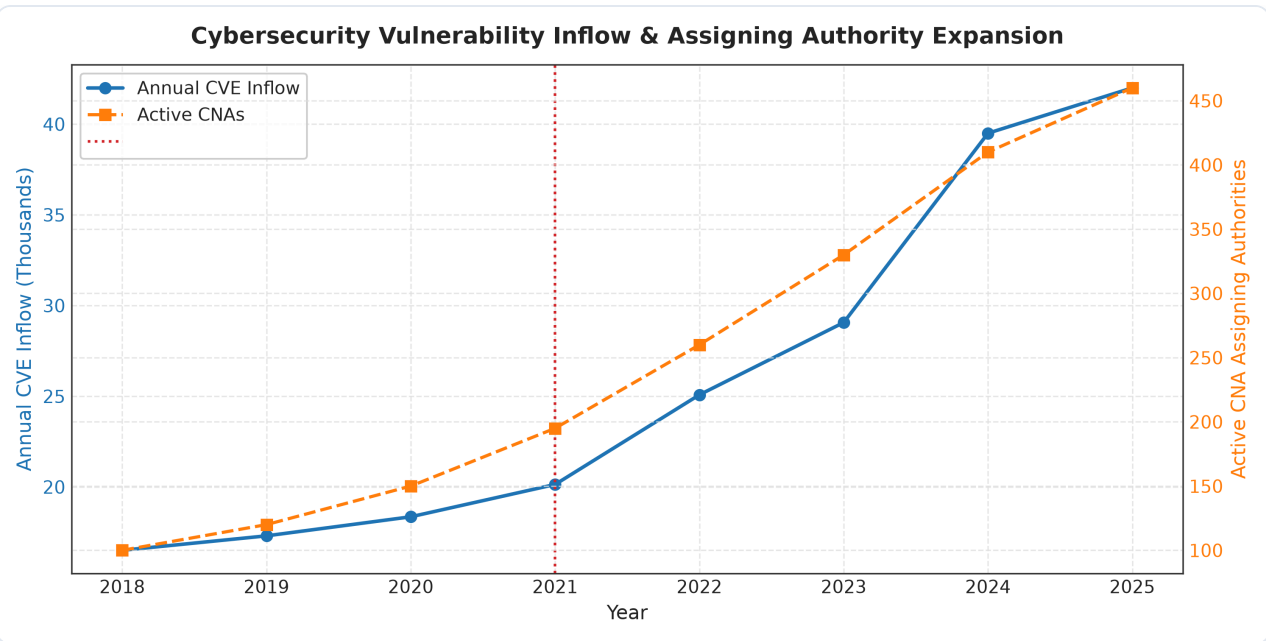


Figure 5.1: Annual CVE Inflow and CNA Assigning Authority Decentralization (2018–2025).

6. Replication Protocol & File Ledger

To replicate this entire research package from scratch:

```
python scripts/00_download_all.py
python scripts/01_validate_sources.py
python scripts/02_process_housing.py
python scripts/03_process_ai_metr.py
python scripts/04_process_anthropic.py
python scripts/05_process_longevity.py
python scripts/06_process_cybersecurity.py
python scripts/07_run_all_models.py
python scripts/08_make_tables.py
python scripts/09_make_figures.py
pytest -q
```